

Oral Consumption of Cranberry Juice Cocktail Inhibits Molecular-Scale Adhesion of Clinical Uropathogenic Escherichia coli

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Cranberry juice cocktail (CJC) has been shown to inhibit the formation of biofilm by uropathogenic *Escherichia coli*. In order to investigate whether the anti-adhesive components could reach the urinary tract after oral consumption of CJC, a volunteer was given 16 oz of either water or CJC. Urine samples were collected at 0, 2, 4, 6, and 8 hours after consumption of a single dose. The ability of compounds in the urine to influence bacterial adhesion was tested for six clinical uropathogenic *E. coli* strains, including four P-fimbriated strains (B37, CFT073, BF1023, and J96) and two strains not expressing P-fimbriae but exhibiting mannose-resistant hemagglutination (B73 and B78). A non-fimbriated strain, HB101, was used as a control. Atomic force microscopy (AFM) was used to measure the adhesion force between a silicon nitride probe and bacteria treated with urine samples. Within 2 hours after CJC consumption, bacteria of the clinical strains treated with the corresponding urine sample demonstrated lower adhesion forces than those treated with urine collected before CJC consumption. The adhesion forces continued decreasing with time after CJC consumption over the 8-hour measurement period. The adhesion forces of bacteria after exposure to urine collected following water consumption did not change. HB101 showed low adhesion forces following both water and CJC consumption, and these did not change over time. The AFM adhesion force measurements were consistent with the results of a hemagglutination assay, confirming that oral consumption of CJC could act against adhesion of uropathogenic *E. coli*.